

GOVERNMENT ENGINEERING COLLEGE – PATAN (022)
COMPUTER SCIENCE & ENGINEERING DEPARTMENT

Subject: Theory of Computation (3160704)

Assignment-2 (CO4)

1. Build a TM that accepts the language PALINDROME over $\{a, b\}^*$. Also trace out the same on the input string abba.
2. Apply the rules and step by step create a Turing Machine to accept $L = \{a^n b^n\}$.
3. Construct PDA for
 $S \rightarrow 0AB$
 $A \rightarrow 1A \mid 1$
 $B \rightarrow 0B \mid 1A \mid 0$
 Trace the string 01011 using PDA.
4. Define and explain the working of Turing Machines and types of Turing Machines.
5. Design a PDA to accept strings with more a's than b's. Also trace out the same on input string abbabaa.
6. Compare Deterministic PDA and Non deterministic PDA.